

Notes: Choi, Kirpalani & Perez (JPE, 2026)

US Public Debt and Safe Asset Market Power

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1 Big picture

- **Question.** Does the US government behave like a price taker when issuing Treasuries, or does it internalize the downward-sloping global demand for US safe assets and exploit market power?
- **Core mechanism.** US public debt provides safety, liquidity, and collateral services to global investors. Because the United States is the dominant supplier of this special asset, additional issuance lowers its convenience yield and raises borrowing costs.
- **Main theoretical result.** If the United States behaves strategically, the convenience yield contains a markup and the equilibrium features underprovision of safe assets relative to a price-taking benchmark.
- **Empirical strategy.** The authors use rotations in the demand curve, measured by high versus low residualized foreign volatility regimes, to distinguish strategic behavior from price-taking behavior.
- **Main empirical result.** Demand for US public debt becomes more inelastic during high foreign-volatility periods. Conduct tests reject the price-taking benchmark in favor of the market-power model.
- **Economic takeaway.** The US safe-asset role is not just passive exorbitant privilege. The paper argues that the United States has and uses safe-asset market power, creating global scarcity and redistributing welfare toward the United States.

2 Data and measurement

2.1 Safe assets and convenience yield

The model and empirics define the convenience yield as the spread between returns on close private substitutes and US Treasury debt. The baseline spread is a weighted average of short- and long-term yield spreads between corporate and Treasury bonds. The quantity variable is privately held gross federal debt relative to US GDP.

Interpretation: The convenience yield is the price of the nonpecuniary services supplied by US debt.

2.2 Demand rotator: foreign volatility regimes

The demand rotator is an indicator for periods when residualized foreign volatility is above its sample median. Foreign volatility is constructed from UK equity-market volatility and residualized using US stock-market volatility and US GDP growth.

Interpretation: A demand rotator is useful because it changes the slope of demand. That rotation is what makes it possible to test conduct rather than merely estimate a downward-sloping demand curve.

2.3 Demand estimation and instruments

The demand estimation uses quarterly data from 1935 to 2019 and controls for the foreign-volatility regime, US volatility, and the Treasury yield-curve slope. The IV strategy instruments debt supply with the US dependency ratio and military spending news shocks.

Interpretation: The instruments are intended to move debt supply without directly moving global safe-asset demand.

2.4 Quantitative calibration moments

The quantitative model is calibrated to match debt quantities, convenience yields, interest rates, external balance-sheet moments, asset home bias, net foreign assets, and US/RoW relative GDP.

Interpretation: The calibration links the empirical conduct estimates to macroeconomic counterfactuals.

3 Models and empirical specifications

3.1 Convenience yield

$$S_t = (r_{K,t} - \delta) - r_t.$$

Interpretation: A positive spread means investors accept a lower Treasury return because Treasuries provide special services.

3.2 US government static monopoly problem

$$\begin{aligned} c_t + a_{t+1} &= w_t + (1 - \delta + r_{K,t})a_t + S_t(b_t^*)b_t^* - x_t(b_t^*). \\ S_t(b_t^*) &= x'_t(b_t^*) - S'_t(b_t^*)b_t^*. \end{aligned}$$

Interpretation: The last term is the monopoly pricing wedge: the government internalizes that more debt lowers the convenience yield.

3.3 Rest-of-world demand and markup

$$\begin{aligned} S_t(b_t^*) &= f'_t(b_t^*), \quad m_t = -\frac{f''_t(b_t^*)}{f'_t(b_t^*)}b_t^* = \frac{1}{\varepsilon_{D,t}}. \\ S_t^{ME} &= \frac{x'_t(b_t^{ME})}{1 - m_t}, \quad S_t^{PT} = x'_t(b_t^{PT}). \end{aligned}$$

Interpretation: A lower demand elasticity implies a higher markup. The price-taking benchmark sets the markup to zero.

3.4 Demand estimation equation

$$S_t = \alpha + \beta \log b_t + \gamma(\log b_t \times z_t) + \delta X_t + \varepsilon_t.$$

Interpretation: The coefficient γ captures the demand rotation across high- and low-volatility regimes.

3.5 Conduct test

$$\log mc_t = \lambda \log b_t + q_t, \quad q_t = \log S_t + \log(1 - ym_t) - \lambda \log b_t,$$

where $y = 1$ for monopoly and $y = 0$ for price-taking.

Interpretation: The correct conduct model should imply a marginal-cost shifter orthogonal to the demand rotator.

4 Identification and empirical design

- **Why price and quantity are not enough.** A downward-sloping demand curve does not prove market power when marginal costs are unobserved.
- **Bresnahan-style identification.** Demand rotations change prices and quantities differently under monopoly and price-taking conduct.
- **Demand rotator validity.** Residualized foreign volatility is designed to capture global flight-to-safety demand rather than US fiscal supply shocks.
- **OLS versus IV.** IV estimates use demographic and military-news instruments; similar conclusions from OLS and IV strengthen the empirical case.
- **Moment-condition logic.** The conduct test compares whether the demand rotator is correlated with model-implied marginal-cost shocks.
- **Remaining limitations.** The test depends on the rotator exclusion restriction, convenience-yield measurement, and the structural mapping from elasticity to markups.

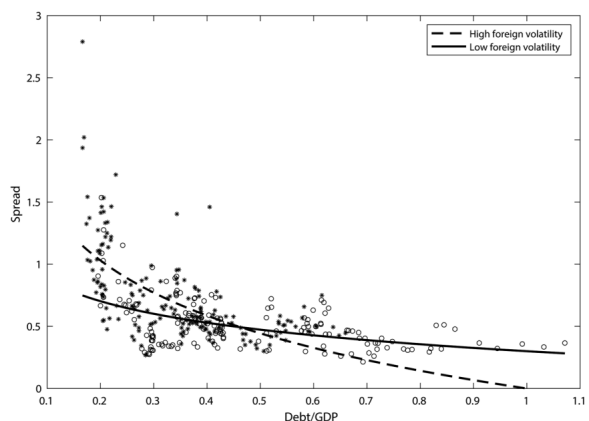
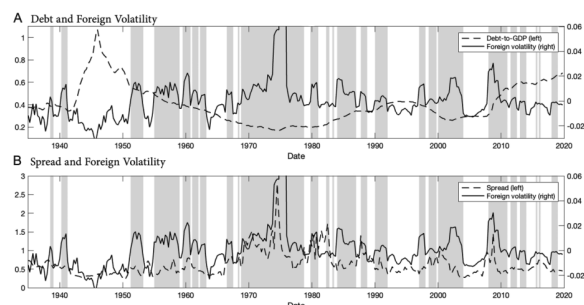
5 Tables and figures

5.1 Figure 1: Evolution of Prices and Quantities of US Public Debt

Read: The figure tracks the convenience yield, Treasury debt relative to GDP, and foreign-volatility regimes over the long sample.

Interpretation: It motivates foreign volatility as a demand rotator and shows episodes when safe-asset demand plausibly becomes more inelastic.

Economic message: High foreign volatility makes the safe-asset demand curve more favorable to the US issuer.



5.2 Figure 2: Safe Asset Demand in Times of High and Low Volatility

Read: The figure plots spreads against debt/GDP separately for high- and low-volatility periods.

Interpretation: The high-volatility demand relationship is steeper, implying lower elasticity and higher markups.

Economic message: This visual is the empirical bridge between safe-asset demand and conduct testing.

5.3 Table 1: Baseline Demand Estimation

Read: The table estimates the demand equation with and without the foreign-volatility interaction, using OLS and IV.

Interpretation: It establishes that demand is more inelastic in high-volatility periods and that implied markups are higher.

Economic message: The table converts safe-asset demand into market-power primitives.

TABLE 1
BASELINE DEMAND ESTIMATION

	OLS (1)	IV (2)	OLS (3)	IV (4)
log(debt/GDP)	-.23*** (.04)	-.16*** (.05)	-.38*** (.03)	-.24*** (.04)
Foreign volatility dummy	-.31*** (.07)	-.12 (.09)	.05* (.03)	.08*** (.03)
Foreign volatility dummy $\times$ log(debt/GDP)	-.34*** (.06)	-.18** (.08)		
US VIX	.02*** (.00)	.02*** (.00)	.02*** (.00)	.02*** (.00)
Yield curve slope	-.06*** (.01)	-.07*** (.01)	-.06*** (.01)	-.07*** (.01)
Constant	.11** (.06)	.17*** (.07)	-.05 (.05)	.05 (.06)
Observations	339	339	339	339
R^2	.52		.49	
Demand elasticity:				
High foreign volatility	1.09	1.82	1.63	2.55
Low foreign volatility	2.68	3.78	1.63	2.55
Markup:				
High foreign volatility	.92	.55	.61	.39
Low foreign volatility	.37	.26	.61	.39

NOTE.—The dependent variable is the weighted average of yield spreads between corporate and Treasury bonds, both measured in percentage units. Foreign volatility dummy is an indicator for whether the residualized foreign volatility measure is above the sample median. The main independent variables of interest are the log of the ratio of the Treasury debt outstanding to US GDP and its interaction with the foreign volatility dummy. Other controls include the slope of the Treasury yield curve (measured as the spread between the 10-year Treasury yield and the 3-month Treasury yield) and a measure of US volatility based on the VIX data. See app. B for a description of the construction of all the variables. The estimation method is OLS for cols. 1 and 3 and IV for cols. 2 and 4. Standard errors are in parentheses.

* Statistically significant at the 10% level.

** Statistically significant at the 5% level.

*** Statistically significant at the 1% level.

TABLE 2
CONDUCT TESTS FOR DIFFERENT COST ELASTICITIES

		Cost Elasticity							
		$\lambda = 0$	$\lambda = 1$	$\lambda = 2$	$\lambda = 3$	$\lambda = 4$	$\lambda = 10$	$\lambda = 20$	$\lambda = 30$
		A. OLS							
T^{RV}		2.69	-2.91	-5.92	-6.99	-7.42	-7.93	-8.02	-8.03
$\hat{p}$		.01	.0	.0	.0	.0	.0	.0	.0
		B. IV							
T^{RV}		-4.06	-6.23	-7.2	-7.72	-8.03	-8.69	-8.94	-9.03
$\hat{p}$		.0	.0	.0	.0	.0	.0	.0	.0

NOTE.—The table shows results of the RV statistical test comparing the fit of the monopoly and price-taking models for different values of the cost elasticity, λ . Values lower than -1.96 reject the price-taking model in favor of the monopoly model. OLS and IV are the methods for estimating the demand for public debt.

5.4 Table 2: Conduct Tests for Different Cost Elasticities

Read: The table reports Rivers-Vuong tests across alternative marginal-cost elasticities.

Interpretation: The test consistently favors the monopoly model because high-volatility spreads are better rationalized by markup movements.

Economic message: This is the paper's main empirical evidence that the United States behaves strategically.

5.5 Table 3: Calibrated Parameters

Read: The table reports preference, benefit, cost, and production parameters.

Interpretation: The calibration determines how convenience benefits and debt issuance costs behave in the quantitative model.

Economic message: These primitives discipline the counterfactual supply and welfare exercises.

Parameter	Description	Value
A. Households and Government		
β	Discount rate	.9886
γ	Risk aversion	2
η	Benefit elasticity	.545
$\bar{\nu}$	Benefit parameter	-5.47
σ_ν	Benefit volatility	.01
ρ_ν	Benefit persistence	.99
λ	Cost elasticity	1
$\bar{\omega}$	Cost parameter	-5.07
σ_ω	Cost volatility	.3
ρ_ω	Cost persistence	.30
B. Firms		
α	Capital share	.3
δ	Depreciation rate	.1
ι	US own capital share	.91
ι^*	RoW own capital share	.79
θ	Capital substitutability	1
$\bar{A}$	US productivity	.82
$\bar{A}^*$	RoW productivity	.93
σ_A	Productivity volatility	.02
ρ_A	Productivity persistence	.95

Moments	Data	Model
A. Targeted Moments		
Mean(interest rate) (%)	.53	.51
Mean(public debt)	.41	.43
Mean(convenience yield) (%)	.62	.62
Var(public debt)	.03	.01
Var(convenience yield)	.086	.07
Corr(public debt, convenience yield)	-.56	-.67
Autocorr(public debt)	.96	.99
Autocorr(convenience yield)	.71	.92
Mean(asset home bias)	.80	.80
Mean(net foreign assets)	-.05	-.07
Mean(US-RoW GDP ratio)	1.10	1.10
B. Untargeted Moments		
Var(current account)	.03	.02
Var(fiscal deficit)	.002	.0002
Var(net foreign assets)	.035	.014
Corr(public debt, net foreign assets)	-.65	-.99
Corr(current account, fiscal deficit)	-.21	-.98

NOTE.—The table reports the moments of different macroeconomic variables for the United States in the data and model. Model moments are averages of stochastic simulations. Data moments are averages of our sample period 1935–2019, unless noted otherwise. Appendix B provides detailed information on the construction of the dataset. Interest rate refers to the interest rate on US public debt. Convenience yield in the model is the difference between the return on US capital and the interest rate on US public debt. Public debt, net foreign assets, current account, and fiscal deficit all correspond to the United States and are expressed as a ratio to GDP. Asset home bias is the share of total US capital used by domestic firms.

5.6 Table 4: Calibration Moments

Read: The table compares targeted and untargeted data moments with model moments.

Interpretation: It checks that the model reproduces key debt, spread, persistence, and external-balance-sheet moments.

Economic message: The counterfactuals are credible only to the extent the model matches these macro-financial moments.

5.7 Table 5: Macroeconomic Distortions due to Market Power

Read: The table compares monopoly and price-taking equilibria, with and without a domestic fringe.

Interpretation: Market power lowers public safe-asset supply and raises the convenience-yield markup.

Economic message: Market power creates global safe-asset scarcity.

TABLE 5
MACROECONOMIC DISTORTIONS DUE TO MARKET POWER

	Baseline Monopoly Model		Model with Domestic Fringe	
	Monopoly Equilibrium (1)	Price-Taking Equilibrium (2)	Monopoly Equilibrium (3)	Price-Taking Equilibrium (4)
US public debt	.43	.62	.41	.64
Private safe assets	NA	NA	.82	.73
Convenience yield (%)	.62	.53	.64	.61
Interest rate on public debt (%)	.51	.61	.54	.57

NOTE.—The table reports the average values of macroeconomic variables from the simulations of different models. Baseline monopoly model refers to the model in which the US government is the sole provider of safe assets. In this model, monopoly equilibrium (col. 1) refers to the equilibrium where the US government acts strategically, and price-taking equilibrium (col. 2) refers to one in which the US government acts as a price taker. The model with domestic fringe refers to the model extension with a private financial sector. In this model, monopoly equilibrium (col. 3) refers to the equilibrium in which the United States acts strategically in the presence of a price-taking domestic fringe, and price-taking equilibrium (col. 4) refers to one in which both the US government and the domestic fringe are price takers. US public debt is expressed as a ratio to US GDP. Convenience yield is the difference between the return on US capital and the interest rate on US public debt. Private safe assets refer to the safe assets issued by the domestic fringe and is expressed as a ratio to US GDP. NA = not applicable.

TABLE 6
WELFARE IMPLICATIONS OF MARKET POWER IN SAFE ASSETS

	Baseline Monopoly Model (%) (1)	Model with Domestic Fringe (%) (2)
US welfare losses:		
No exorbitant privilege	-.20	-.32
No market power in safe assets	-.09	-.09
RoW welfare gains:		
No market power in safe assets	+.12	+.11

NOTE.—The baseline monopoly model (col. 1) refers to the model in which the US government is the sole provider of safe assets. In this model, no exorbitant privilege refers to the US welfare losses of transitioning from the baseline monopoly equilibrium to an economy in which the RoW does not enjoy a nonpecuniary value of holding US public debt, and no market power in safe asset refers to the welfare effects of transitioning from the baseline monopoly equilibrium in which the US government acts strategically to one in which it acts as a price taker. The model with domestic fringe (col. 2) refers to the model extension with a private financial sector. In this model, no exorbitant privilege refers to the US welfare losses of transitioning from the equilibrium in which the US government acts strategically in the presence of a price-taking domestic fringe to an economy in which the RoW does not enjoy a nonpecuniary value of holding any type of safe asset, and no market power in safe asset refers to the welfare effects of transitioning from the equilibrium in which the US government acts strategically in the presence of a price-taking domestic fringe to one in which both the US government and the domestic fringe are price takers. Welfare changes are expressed in permanent consumption equivalence terms considering the transition.

5.8 Table 6: Welfare Implications of Market Power in Safe Assets

Read: The table reports welfare losses or gains in permanent consumption-equivalent terms.

Interpretation: Removing market power reduces US welfare and raises RoW welfare.

Economic message: The convenience yield is an international surplus-sharing mechanism.

5.9 Table 7: Effects of Increasing Competition

Read: The table studies Cournot competition with two or three safe-asset issuers.

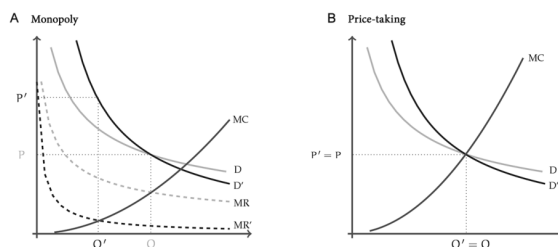
Interpretation: Competition increases global safe-asset supply, lowers convenience yields, raises US borrowing costs, and reduces US welfare.

Economic message: Alternative safe-asset suppliers would reduce US safe-asset market power.

TABLE 7
EFFECTS OF INCREASING COMPETITION

	Monopoly	Cournot	
		$N = 2$	$N = 3$
		A. Steady-State Variables	
Total safe assets	.43	.81	1.12
US public debt	.43	.41	.37
Convenience yield (%)	.62	.47	.40
Interest rate on public debt (%)	.51	.67	.74
B. Welfare			
US welfare (%)		-.13	-.18
RoW welfare (%)		+.18	+.29

NOTE.—The table shows key macroeconomic variables under different competition arrangements. Panel A shows model moments as averages of stochastic simulations. Panel B reports the welfare change expressed in permanent consumption equivalence terms of transitioning from the monopoly equilibrium to different economies. Total safe assets are the sum of public debt issued by every country. US public debt is public debt of only the United States. Both of these are expressed as a ratio of US GDP. Convenience yield is the spread between the returns on US capital and US public debt. Interest on public debt is the interest rate on US public debt.



5.10 Appendix Figure C1: Demand Rotations to Identify US Government Conduct

Read: The figure illustrates why rotations in demand help distinguish monopoly from price-taking behavior.

Interpretation: It is the graphical intuition behind the conduct test.

Economic message: Identification borrows from industrial organization.

5.11 Appendix Figure C2 and Table C1: Fiscal Supply Instruments and Debt-Volatility Comovement

Read: Figure C2 plots fiscal supply instruments and Table C1 reports comovement with volatility.

Interpretation: These exhibits support the instrument and rotator discussion.

Economic message: They help separate debt-supply shifts from safe-asset demand rotations.

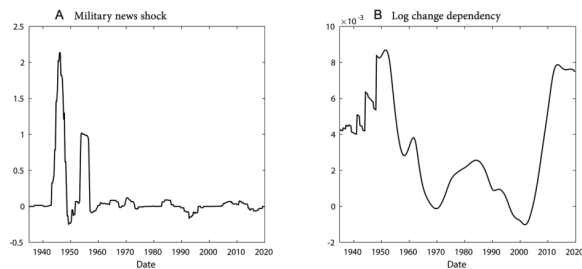


FIG. C3.—Cournot transition. The figure shows the path of macroeconomic variables in response to increasing the number of safe asset issuers to N from the steady state of an economy with a single safe asset issuer. RoW consumption is consumption in the RoW. Spread is the difference between the net returns on capital and the returns on safe assets.

TABLE C1
DEBT-TO-GDP RATIO AND VOLATILITY COMOVEMENT

	UK Volatility	UK Volatility Indicator
Debt-to-GDP ratio	-.39	-.34
Detrended log real debt	-.30	-.27

NOTE.—We report correlations in the table. UK volatility is a 1935–2019 historical series of the annualized rolling 4-quarter standard deviation of the weekly log stock returns on the MSCI United Kingdom Index, predicted by regressing the series on the annualized

5.12 Appendix Figure C3: Cournot Transition

Read: The figure plots transition paths when the number of safe-asset issuers increases.

Interpretation: It shows that competition affects prices, quantities, and consumption dynamically.

Economic message: Safe-asset competition has transition dynamics, not only steady-state consequences.

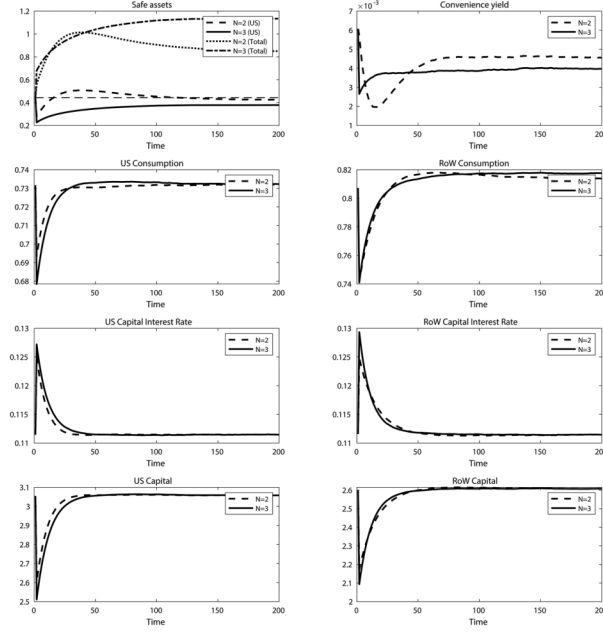


Figure 1: Cournot transition

6 Short summary

- The paper studies whether the US government exploits market power as the dominant supplier of global safe assets.
- US public debt provides nonpecuniary safety, liquidity, and collateral services.
- A strategic issuer restricts safe-asset supply because additional issuance lowers the convenience yield.
- Foreign volatility is used as a demand rotator because flight-to-safety episodes make Treasury demand less elastic.
- Conduct tests reject price-taking behavior in favor of the market-power model.
- The calibrated model implies substantial underprovision of safe assets and a markup component in the convenience yield.
- Competition from other safe-asset issuers increases global supply and reduces US welfare.

7 Conclusion

7.1 Core conclusion

Choi, Kirpalani, and Perez argue that the US government is not merely a passive borrower benefiting from high global demand for Treasuries. It faces a downward-sloping demand curve and behaves as a strategic safe-asset supplier.

7.2 Economic mechanism

Additional Treasury issuance lowers the convenience yield on all inframarginal debt. A strategic government therefore preserves scarcity when demand is inelastic, creating a markup in the convenience yield.

7.3 Evidence chain

The paper estimates demand elasticities across foreign-volatility regimes, uses demand rotations to test conduct, and calibrates a macro model to quantify the resulting distortions.

7.4 Broader implications

Safe-asset market power affects fiscal capacity, global collateral scarcity, reserve-currency competition, and welfare distribution between the United States and the rest of the world.

7.5 Caveats

The conclusion depends on the demand-rotator exclusion restriction, convenience-yield measurement, and the mapping from demand elasticity to markups.

7.6 Final takeaway

The convenience yield is partly a scarcity rent. US public debt market power helps explain both the low borrowing cost of the United States and the global underprovision of safe assets.